

Sunny Zhang

(289) 885-1291 ◇ sunzhang0510@gmail.com ◇ LinkedIn: <https://www.linkedin.com/in/sunnyzhang08/>

EDUCATION

University of Waterloo, Faculty of Engineering
Bachelor of Applied Science in Mechatronics Engineering

Expected graduation date: 2031

- Recipient, Guo Family Entrance Scholarship (\$5,000) and President's Scholarship of Distinction (\$2,000)
- High School Average: 98%

Relevant Coursework: Chemistry for Engineers, Linear Algebra for Engineering, Calculus 1 for Engineering, Digital Computation, Mechatronics Engineering

EXPERIENCE

Ace Manufacturing Inc.
Summer Intern

June 2026 - September 2026

- Executed drawing revisions and finalized design specifications in AutoCAD for a burn-off oven facility addition, matching requirements for S&C Electric Canada.
- Designed pump skids with covers for ControlChem Canada Ltd., platforms and stairs, and flat patterns for interlocking steel panel walls using AutoCAD.
- Shadowed the design and physical construction of a large scale industrial coating removal system over two months
- Fabricated metal components and pipes for assemblies using lathes, milling machines, and metal cutting bandsaws
- Audited pipe fitting dimensions, types, and inventory counts for order fulfillment

PROJECTS

2D Mapping Vehicle

September 2026-Present

- Engineered an Arduino-based differential vehicle utilizing time-of-flight sensors, quadrature wheel encoders, and IMU sensors for precise 2-D position tracking and mapping
- Designed and tested a multi-layered chassis in SolidWorks
- Programmed embedded control architecture in C++ to process differential-drive kinematics and calculate real-time coordinates
- Developed a live Python plotting script to parse serial data and dynamically visualize the vehicle's mapped environment and detected boundaries
- Drafted wiring diagrams in Fritzing

AI Dashcam Prototype at SHAD Canada

June 2025 - July 2025

- Conceptualized the system architecture and business case for an AI-powered dash-cam designed to automate road damage detection
- Pitched prototype concept and market viability to a panel of industry judges, winning 'Best Presentation' and 'Best Prototype' at the SHAD program

Professional Portfolio Website

September 2026 - present

- NOTE: THIS IS A PROJECT I PLAN ON DOING SOON
- Designed, styled, and deployed a personal website from scratch using HTML, CSS, and JavaScript, implementing a responsive layout and custom styling
- Built an interactive dark/light mode toggle using JavaScript to add dynamic functionality to the site
- Managed source code and version history using Git, hosting the repository on GitHub with a documented README
- Deployed the site live via GitHub Pages, making it publicly accessible at [placeholder]

Robot Vacuum (Roomba)

April 2026 - May 2026

- Designed autonomous vacuum vehicle integrating a Raspberry Pi Pico W with HC-SR04 ultrasonic sensors and a MPU6050 gyroscope for spatial awareness
- Programmed and debugged closed loop MicroPython code in VSC for obstacle avoidance and course correction
- Designed chassis in Onshape and 3D printed all parts, iterating on the design to avoid collisions and protect hardware components, capable of making a full stop and turning around within 3 cm of an obstacle.

SKILLS

Software: Java, Python, C++, Arduino IDE, HTML, CSS, JavaScript, Visual Studio Code

Hardware: Arduino, Raspberry Pi, Embedded systems

CAD/Design: SolidWorks, AutoCAD, Onshape, Fritzing